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Exploring the Applicability of Three Popular Models for Predicting the Received Field Strength of HF Sky-Wave Propagation and Communication

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Abstract High-frequency (HF) radio signals rely on ionospheric reflection and refraction to enable long-range and transcontinental transmission, making them essential for military command, emergency response, and strategic communications. The received field strength is a key parameter for quantifying propagation conditions and optimizing HF circuit design and application. To seek the optimal model and its improvement method, therefore, this study evaluates the prediction performance of three models (REC533, ICEPAC, and VOACAP) using the Consultative Committee on International Radio (CCIR) data set and the International Telecommunication Union (ITU) recommended standardized procedure. The assessment covers multiple dimensions, including time, distance, operating frequency, season, solar activity, and the midpoint position of the path. The results show that REC533 achieves the best overall performance, exhibiting a near-zero bias of 0.28 dB(μV/m) and the lowest standard deviation of prediction errors, 9.81 dB(μV/m), together with consistently high stability across all propagation scenarios. In contrast, ICEPAC and VOACAP exhibit systematic underestimation of approximately 4 dB(μV/m), with substantially larger dispersion exceeding 15 dB (μV/m), particularly for long-distance circuits, at high frequencies, and in low-latitude regions. These findings clarify the applicability and limitations of the three prediction models, provide quantitative guidance for HF spectrum planning, circuit design, and emergency communication support, and offer directions for future model refinement.

Plain Language Summary High-frequency (HF) radio waves can travel long distances by reflecting from the ionosphere, making them essential for communication in remote regions, disaster response, aviation, and maritime operations. The received field strength, however, dynamic variations with time of day, distance, season, solar activity, and location on Earth. Accurate prediction of received strength is therefore critical for ensuring reliable HF communication. To seek the optimal model and its improvement method, therefore, this study evaluates three widely used prediction models—REC533, ICEPAC, and VOACAP—by comparing them with a large international data set of real HF measurements collected over many years. We examine how well the models perform under different conditions, such as long or short transmission paths, low or high solar activity, and different geomagnetic regions. We find that REC533 shows the highest prediction accuracy and stability, characterized by a near-zero bias of 0.28 dB(μV/m) and the lowest dispersion of 9.81 dB(μV/m). In contrast, ICEPAC and VOACAP tend to underestimate field strength by approximately 4 dB(μV/m) and exhibit dispersion exceeding 15 dB(μV/m), particularly for long-distance circuits, at high frequencies, and in low-latitude regions. These results clarify the applicability of the three models and support improved HF spectrum planning, circuit design, and emergency communication.

Zhanfeng Zhu and Qiao Yu contributed equally to this work and are co-first authors.

Key Points:

- The prediction performance of REC533, ICEPAC, and VOACAP has been evaluated based on the global HF field strength data
- REC533 provides the most accurate and stable HF field strength predictions across times, distances, frequencies, and latitudes
- ICEPAC and VOACAP consistently underestimate field strength, especially for long paths, high frequencies, and low latitude regions

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1. Introduction

High-frequency (HF), as defined by the International Telecommunication Union (ITU) for radiocommunication and radar services, refers to the 3–30 MHz frequency band (International Telecommunication Union, 2015), traditionally supporting transmission ranges up to above 10,000 km using sky-wave propagation (Wang, Shi, et al., 2022). HF sky-wave communication exploits one or multiple reflections and refractions within the ionosphere, enabling long-distance and even global transmission (Maslin, 2017; Wang et al., 2026). This communication mode also offers advantages such as high mobility, flexible networking, and substantial survivability. Its autonomous communication capability and survivability surpass those of other communication methods, while effectively avoiding issues like high costs and sovereignty concerns (Wang et al., 2018). Despite the emergence of

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new wireless communication technologies such as microwave and satellite systems, HF communication continues to advance (Lu, 2005). It remains indispensable in military, governmental, commercial, and emergency response applications (Johnson et al., 1997). In addition, this mode is also widely used for beyond-line-of-sight information transmission in aeronautical services, maritime operations, and remote areas where other system infrastructures are unavailable. HF sky-wave propagation is strongly controlled by ionospheric variability, which is driven by solar radiation, geomagnetic activity, solar flares, energetic particle precipitation, and seasonal changes (Wang, Yang, & An, 2022). These space-weather-related disturbances can modify ionospheric electron density, absorption, MUF, and propagation-mode availability, thereby affecting the reliability of HF communication links (Yu, Wang, Yang et al., 2026). Therefore, evaluating the performance of commonly used HF prediction models is important for space-weather-related communication assessment and operational decision-making (Frissell et al., 2019).

In HF sky-wave communication, one of the core requirements is to accurately quantify the propagation characteristics. The selection and optimization of prediction methods for received field strength, a key metric describing the process, have long been a core focus of research in this field. Barabashov et al. (2006) proposed a three-dimensional spatio-temporal prediction model based on a structural physics theory to characterize the spatial-temporal features of short-range sky-wave propagation, and subsequently proposed a method for accurately predicting the characteristics of HF sky-wave propagation in magnetically active and spatially inhomogeneous ionospheres (Barabashov et al., 2009). He and Zhao (2009) proposed a method to calculate ionospheric absorption attenuation. They analyzed the relationship between the absorption attenuation and the frequency and elevation angle of the sky-wave using simulations. Walden (2012) analyzed UK HF beacon measurements at 5.290 MHz and found strong seasonal prediction biases in VOACAP and ASAPS. Summer and winter deviations were linked to sporadic E and enhanced ionospheric absorption. Kotova et al. (2015) combined the global self-consistent model of the thermosphere, ionosphere, and protonosphere with sky-wave propagation to calculate propagation loss in the three-dimensional inhomogeneous low-latitude ionosphere during electromagnetic storms. Siskind et al. (2017) showed that a WACCM-based model improves D- and E-region electron densities. Persistent low-latitude D-region deficits introduce uncertainty in HF absorption predictions. Wang et al. (2019) proposed a parabolic equation based on the propagation of sky-waves, which was used to calculate the attenuation of HF circuits. Pandey et al. (2022) simulated parameters such as received power using Faraday's rotational inversion phenomenon by modeling the polarization characteristics of sky-waves. Ge et al. (2024) simulated several circuits, including those from Qingdao, China, to Tromsø, Norway, and analyzed the predictions of signal power.

Currently, the most common field strength prediction models in HF communication are Recommendation ITU-R P.533 (REC533), Ionospheric Communications Enhanced Profile Analysis and Circuit (ICEPAC), and Voice of America Coverage Analysis Program (VOACAP). REC533 is derived from the Radiocommunication Sector of Recommendation ITU-R P.533, which provides a prediction method for the median field strength and system reliability for both HF analog and digital modulation systems (International Telecommunication Union, 2019). This method originates from the field strength prediction method documented in the Consultative Committee on International Radio (CCIR) Report 894. Its foundation lies in an HF sky-wave propagation prediction program developed by CCIR Interim Working Group 6/12 for HF broadcasting service planning, which was improved by Working Party 6A, and subsequently established as an internationally accepted model (Goodman & Reilly, 1988). It has undergone more than 10 revisions, and the current version is REC 533–14. Both ICEPAC and VOACAP are derived from the empirical Ionospheric Communications Analysis and Prediction Program (IONCAP), which is no longer in operational use. ICEPAC was developed with support from the U.S. Air Force and the National Security Agency as an extension of IONCAP, forming an enhanced HF propagation prediction model. It incorporates modified representations of the ionospheric structure in polar regions and at low latitudes (Stewart, 2008). Voice of America took over refining the IONCAP code. To maintain the integrity of IONCAP, a final review was conducted by Donald L. Lucas, its original author. These efforts culminated in the development of VOACAP (Lane, 2001).

Although REC533, ICEPAC, and VOACAP have been widely used in HF propagation prediction and operational planning, their relative applicability for received field-strength prediction has not been comprehensively reassessed using a unified global measurement data set and a consistent evaluation procedure. In addition, the three models have different assumptions regarding ionospheric representation, propagation loss estimation, and empirical correction, but their practical impacts on received field-strength prediction under different propagation

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conditions remain insufficiently quantified. Therefore, clarifying their performance can provide useful guidance for future engineering applications, such as HF link planning, frequency management, communication-availability assessment, and the selection of suitable prediction models under different propagation conditions.

To assess their applicability and accuracy under varying propagation conditions, this study performs a systematic comparison and validation. Using global field strength measurements and the ITU standardized procedure, the performance of the three prediction models of HF sky-wave field strength is evaluated across time, distance, frequency, and related factors. The structure of this study is as follows. Section 2 introduces the origins and core modeling principles of REC533, ICEPAC, and VOACAP. Section 3 describes the data set and model-evaluation method. Section 4 presents the analysis and discussion, including the overall statistical comparison, circuit-specific validation, field-strength coverage analysis, and physical interpretation of the model differences. Section 5 summarizes the main conclusions and implications of this study.

2. Model Descriptions

In this section, we first introduce the principles of three popular prediction models (REC533, ICEPAC, and VOACAP) for HF sky-wave field strength, providing a theoretical basis for subsequent model performance validation and comparison.

2.1. REC533

REC533 provides a systematic method for predicting the performance of HF circuits under both analog and digital modulation (Yu, Wang, Shi et al., 2026). It accounts for the time-varying characteristics of the ionosphere, the signal-to-noise ratio, channel time spreading, and frequency dispersion. These elements form a unified theoretical framework for HF communication circuit planning and performance evaluation (International Telecommunication Union, 2019).

This model uses different propagation calculation schemes based on path length and ionospheric layer type. When the path distance is less than 7,000 km, the model carries out hop-by-hop analysis based on the ray path geometry theory; when the path is more than 9,000 km, the combined empirical formulas obtained from the regression of the measured data are used; and in the transition interval between 7,000 and 9,000 km, the continuous connection of the two methods is realized by smooth interpolation.

In the field strength prediction component, REC533 defines the median sky-wave field strength E_w as a function of several factors:

$$E_w = 136.6 + P_t + G_t + 20\log(f) - L_b, \tag{1}$$

where P_t is the transmitter power, G_t is the gain of the transmitting antenna at azimuth and elevation angle, and f is the operating frequency. L_b denotes the total transmission loss along the path, expressed as

$$L_b = 32.45 + 20\log(f) + 20\log(p') + L_i + L_m + L_g + L_h + L_z, \tag{2}$$

where L_i is the ionospheric absorption loss, L_m is the additional loss above maximum usable frequency (MUF), L_g is the ground reflection loss, L_h is the auroral and spurious signal loss, and L_z is the empirical correction term.

In the case of multimode propagation, the model uses a power superposition to synthesize the median field strength:

$$E_s = 10\log\left(\sum 10^{(E_w/10)}\right). \tag{3}$$

For long-distance propagation (distance > 7,000 km), the field strength is corrected by the measured empirical formula:

$$E_{tl} = E_0\left[1 - \frac{(f_M + f_H)^2 (f_L + f_H)^2}{(f_M + f_H)^2 + (f_L + f_H)^2}\right] - 30.0 + P_t + G_{tl} + G_{ap} - L_y, \tag{4}$$

where f_M and f_L represent MUF and lowest usable frequency (LUF), f_H is the electron slew frequency, and G_{ap} is the geometric correction factor for the Earth.

In summary, REC533 provides a continuous characterization of propagation behavior from short to long range paths and across the E and F layers. The output from the model, combined with the noise model proposed by Recommendation ITU-R P.372 (International Telecommunication Union, 2024), can be further used to calculate signal-to-noise ratio and basic circuit reliability. These parameters provide a quantitative basis for HF circuit design under different seasons, solar-activity conditions, and geographic regions.

2.2. ICEPAC

ICEPAC was developed by the Institute for Telecommunication Sciences of the National Telecommunications and Information Administration as an extension of IONCAP. It is an HF propagation prediction model that incorporates polar-region structure, ionospheric conductivity, and electron-density distributions. Its general framework is similar to that of IONCAP, but with systematic improvements in ionospheric partition modeling, path loss calculation, and noise estimation (Stewart, 2008).

ICEPAC adopts the ionospheric conductivity and electron density description to model the ionosphere. It divides the ionosphere into typical regions, including the subauroral trough, auroral zone, and polar cap. ICEPAC uses Chapman layer functions to construct electron-density profiles. This approach enables continuous three-dimensional modeling of ionospheric electron density and conductivity across different latitudes and times. This method provides a more realistic representation of ionospheric structure in both the horizontal and vertical dimensions, and significantly improves prediction accuracy for polar and high-latitude circuits (Lane, 2005).

For short-distance propagation (distance < 10,000 km), ICEPAC uses the hop-ray model. ICEPAC formulates the field-strength prediction as a transmission-loss calculation. It first determines the available propagation modes from the electron-density profile and path geometry. The system loss for a selected mode can be expressed as

$$L_b = L_{\text{bf}} + L_i + L_g - (G_t + G_r) \tag{5}$$

where L_b is the system loss, L_{bf} is the basic free-space transmission loss, L_i is the ionospheric-loss term, L_g is the ground-reflection loss, and G_t and G_r are the transmitting and receiving antenna gains in the corresponding propagation direction.

After the system loss is obtained, the sky-wave field strength is calculated as

$$E = 107.2 + 20 \log f + G_t + P - L_b \tag{6}$$

where E is the predicted sky-wave field strength, P is the transmitter power, and G_t is the transmitting antenna gain.

For long-distance propagation (distance >10,000 km), the ICEPAC uses the forward-scatter model. It estimates the predicted field strength through a long-distance system-loss calculation. The total long-distance system loss can be written as

$$L_{\text{sys}} = L_{\text{bf}} - C_H + L_{\text{launch}} + L_{\text{path}} + L_{\text{intercept}} + L_G + Y_P \tag{7}$$

where L_{bf} is the basic free-space loss, C_H is the horizontal convergence gain, L_{launch} is the sky-wave launch loss near the transmitter, L_{path} is the loss accumulated between the launch and intercept areas, $L_{\text{intercept}}$ is the sky-wave intercept loss near the receiver, L_G is the ground-related loss, and Y_P is the empirical system-loss adjustment factor.

The predicted long-distance field strength is then defined

$$E = 107.2 + 20 \log f + G_t + P - L_{sys}, \tag{8}$$

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where E is the predicted sky-wave field strength, P is the transmitter power, and G_t is the transmitting antenna gain.

2.3. VOACAP

Derived from IONCAP, VOACAP was developed to simplify the original model, add regional coverage and mapping functions, correct coding deficiencies, and incorporate more accurate radio-noise prediction algorithms (Teters & Lloyd, 1968; Yu, Li et al., 2026). It integrates mature ionospheric theory with extensive measurement data, including observations from the U.S. Army's global ionosonde network dating back to 1944 and over-the-range radar measurements collected by the U.S. Navy in the 1970s. VOACAP is based on the Friis transmission equation. This equation relates the receiver power to the transmitter power, accounting for the gains of the transmitting and receiving antennas as well as transmission losses. The comprehensiveness of its propagation model, combined with a statistical database over time and frequency, makes it one of the most accurate models for HF system performance prediction (Lane, 2001).

Since VOACAP and ICEPAC are both derived from the IONCAP framework, they use similar field-strength prediction procedures, with the ray-hop model for short-distance propagation and the scatter model for long-distance propagation. Their field-strength prediction follows a similar signal-power and transmission-loss calculation structure. Therefore, the field strength can also be obtained through a loss-to-field-strength relationship similar to that described for ICEPAC. To avoid repetition, the mathematical form is not repeated here.

2.4. Model-Specific Differences Among REC533, ICEPAC, and VOACAP

Although REC533, ICEPAC, and VOACAP are all used for HF sky-wave field-strength prediction, their model structures differ in terms of ionospheric representation, propagation-mode treatment, and long-distance propagation modeling (International Telecommunication Union, 2019; Lane, 2001; Stewart, 2008). In this study, ICEPAC and VOACAP are regarded as IONCAP-derived models, whereas REC533 follows the ITU-R standardized prediction procedure.

The three models use similar basic circuit inputs, including transmitter and receiver locations, path distance, operating frequency, month, hour, solar-activity index, transmitter power, and antenna parameters. These inputs define the propagation path, temporal condition, and radio-system configuration for field-strength prediction.

For short-distance propagation, both REC533 and the IONCAP-derived models consider multiple ionospheric propagation modes. However, the way in which these modes are generated and used is different. REC533 does not explicitly construct a full electron-density profile along the propagation path. Instead, it uses monthly median ionospheric characteristics, such as foE, foF2, and M(3000)F2, evaluated at prescribed control points to estimate the basic MUF, propagation modes, and transmission losses. Candidate E− and F2-layer modes are determined mainly from path geometry, hop length, and control-point ionospheric parameters (International Telecommunication Union, 2019). The median sky-wave field strength is then calculated using an explicitly decomposed transmission-loss formulation.

In contrast, the IONCAP-derived models use an electron-density-profile-based ray-hop framework for short-path prediction. Ionospheric control points from worldwide maps of critical frequency are used to estimate the electron density at different ionospheric layers, and both low-ray and high-ray paths can be computed for different layer-hop numbers. The signal power assigned to different ray paths is then used to obtain the system-performance prediction (Stewart, 2008). Therefore, compared with REC533, the IONCAP-derived short-path framework is more strongly coupled with the assumed electron-density profile, ray geometry, takeoff and arrival angles, antenna gains, and mode-dependent signal-power distribution.

For long-distance propagation, the difference between REC533 and the IONCAP-derived models becomes more evident. REC533 uses the ITU-R standardized long-distance treatment, in which ray-path analysis is applied to shorter paths and a composite-mode empirical formulation is applied to long-distance paths, with a transition region between the two treatments (International Telecommunication Union, 2019). This provides a relatively continuous treatment of field-strength variation with distance.

In contrast, IONCAP-derived models use a long-path forward-scatter model for paths greater than 10,000 km. The forward-scatter model mainly uses ionospheric control points close to the transmitting and receiving terminals

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and does not fully account for the intervening control points along the path (Stewart, 2008). As a result, IONCAP-derived models may produce different attenuation behavior from REC533 over long-distance circuits.

The difference between ICEPAC and VOACAP is mainly related to their ionospheric representations and model-specific modifications. Both models originate from the IONCAP framework and share similar empirical signal-power and system-performance concepts. However, VOACAP was developed mainly to improve the operational implementation of IONCAP, including user accessibility, area-coverage prediction and mapping, correction of program errors, and improved radio-noise prediction (Lane, 2005). ICEPAC, in contrast, introduces substantial changes to the ionospheric structure. It incorporates the ICED-based electron-density and conductivity profile, Chapman-layer representation, and separate treatments of the subauroral trough, auroral zone, and polar cap (Lane, 2005).

In addition, while ICEPAC uses the abrupt transition between short path and long path at 10,000 km. VOACAP uses a smoothing function between the short path (ray hop) and long (forward scatter) models as a function of distance between 7,000 and 10,000 km (Lane, 2005). Previous evaluation has reported that, for paths exceeding 7,000 km, ICEPAC may predict regions with no received signal followed by stronger signal returns at greater distances, indicating possible discontinuity in long-distance coverage behavior.

3. Data and Method

This section introduces the data set and evaluation method used in this study. The CCIR Databank D1 data set, data-selection criteria, and statistical indicators are described. Key concepts used in the analysis, including short-path propagation and path midpoint, are also clarified to provide a consistent basis for model comparison.

3.1. Data Set

Regarding the observations, we use the CCIR Databank D1 data set, which is the result of decades of international cooperation in the field of HF sky-wave field strength estimation. It is an internationally recognized HF sky-wave field-strength database compiled through long-term international cooperation and widely used for the development and validation of HF propagation prediction methods (Bradley, 1998). The data set includes 181 communication circuits among 68 stations distributed worldwide. The information includes the monthly median of the hourly measured field strength, the frequency of the HF signals, the names and latitude/longitude coordinates of the transmitting and receiving stations, and the monthly sunspot number. The recorded sky-wave field strengths have been normalized to 1 kW of equivalent isotropically radiated power. The data set covers January 1964 to November 1985. This interval includes the declining end of Solar Cycle 19 (approximately April 1954–October 1964), Solar Cycle 20 (approximately October 1964–March 1976), and most of Solar Cycle 21 (approximately March 1976–September 1986). The CCIR Databank D1 consists mainly of short-path propagation cases. Specifically, 1,493 records correspond to short-path propagation, while only 121 records correspond to long-path propagation. Here, the short path denotes the shorter great-circle route between the two stations, whereas the long path denotes the complementary great-circle route in the opposite direction around the Earth. In CCIR Databank D1, the short-path and long-path records are clearly separated in propagation distance. The longest short-path distance is 16,448 km, whereas the shortest long-path distance is 23,153 km. This large distance gap indicates that the long-path records correspond to substantially different propagation geometries, ionospheric control regions, local-time conditions along the path, and possible propagation mechanisms. Therefore, mixing short-path and long-path records in the same statistical evaluation may introduce additional uncertainty and make it more difficult to attribute the prediction errors to the models themselves. For this reason, only short-path records were retained in this study to maintain a consistent propagation-geometry basis for comparing REC533, ICEPAC, and VOACAP. The operating frequency range of the selected CCIR Databank D1 records is 2.5–25.8 MHz. This range corresponds to the frequency coverage of the selected valid observation records, and it also falls within the applicable frequency range of HF propagation prediction models such as REC533, ICEPAC, and VOACAP. Each record contains the basic parameters required for field strength prediction, which can be used for subsequent model evaluation. Figure 1 provides an overview of the measurement circuits, showing the locations of all transmitting and receiving stations together with the corresponding circuits used in the analysis. Table 1 summarizes the transmitting and receiving stations, along with the path distances associated with each circuit. The latitudes listed in Table 1 are geographic station latitudes, whereas the latitude used for geomagnetic-latitude classification is the path-midpoint geomagnetic latitude.

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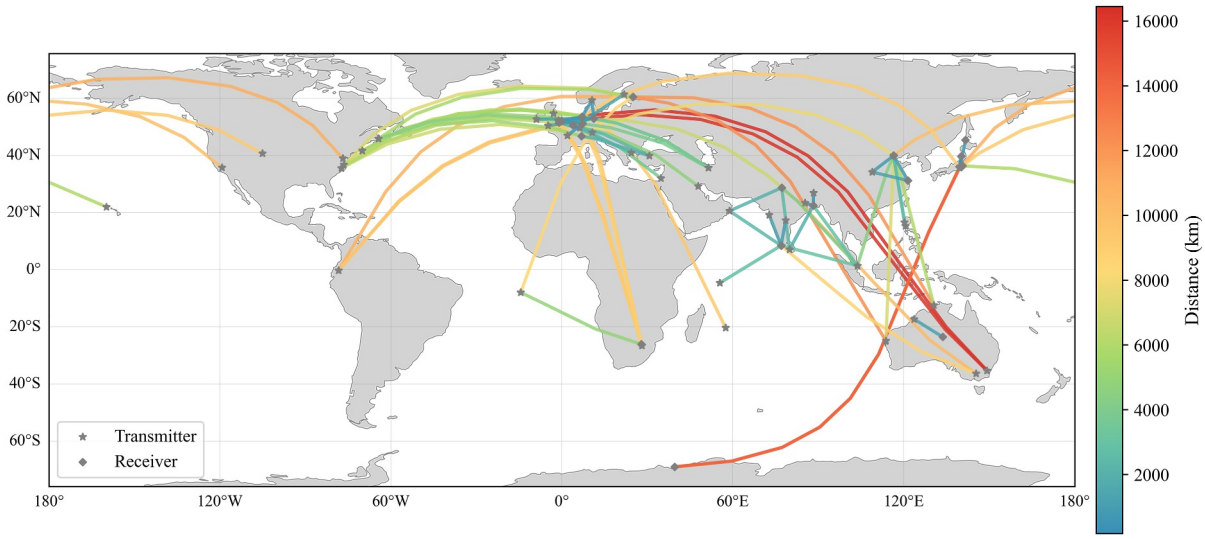


Figure 1. The transmitting-receiving position of measurement circuits.

3.2. Method for Model Evaluation

Comparison of REC533, ICEPAC, and VOACAP was performed with reference to the standardized procedure published by ITU for comparison of predicted and observed HF sky-wave field strengths (International Telecommunication Union, 1997). The comparison metrics include the mean difference (MD) and the standard deviation of prediction errors (SD). For each valid record, the observed value m_i is the monthly median of the hourly measured field strength for a given circuit, month, hour, operating frequency, and sunspot-number condition. The corresponding predicted value p_i is the monthly median field strength calculated by each model under the same input condition. The prediction error is defined as

$$\varepsilon_i = p_i - m_i \tag{9}$$

The MD is calculated as

$$\mathrm{MD} = \frac{1}{n}\sum_{i=1}^{n}\varepsilon_i = \frac{1}{n}\sum_{i=1}^{n}(p_i - m_i), \tag{10}$$

where n is the number of valid paired predicted and observed records. MD represents the systematic prediction bias. A positive MD indicates overestimation, whereas a negative MD indicates underestimation.

The SD used in this study is the standard deviation of the prediction errors, rather than the standard deviation of the measured or predicted field-strength values:

$$\mathrm{SD} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(\varepsilon_i - \mathrm{MD})^2} \tag{11}$$

In accordance with the standardized evaluation procedure recommended in ITU-R P.1148 (International Telecommunication Union, 1997), MD and SD were adopted as the statistical indicators for the overall performance analysis in Section 4.1.1. MD quantifies the systematic bias between predicted and observed monthly median field strengths, whereas SD characterizes the dispersion of the prediction errors. To more clearly characterize the error distributions, we have further visualized the overall prediction-error samples using a combined boxplot–violin plot in Section 4.1.1, supporting the use of MD and SD.

According to Recommendation ITU-R P.1148 (International Telecommunication Union, 1997), the subset used in the comparison was divided into different categories. The division scheme is adapted to the specific research

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Table 1
Transmitting and Receiving Stations, Geographic Positions, and Corresponding Path Distances

ID	TX	TX Geo. Lat. (°)	TX Geo. Lon. (°)	RX	RX Geo. Lat. (°)	RX Geo.Lon. (°)	Dist. (km)
1	Luxemburg	49.40°N	6.19°E	Bockhacken	51.07°N	7.16°E	175
2	Ranchi	23.24°N	85.22°E	Calcutta	22.27°N	88.18°E	319
3	Sanwa	36.11°N	139.51°E	Akita	39.44°N	140.08°E	396
4	Koganei	35.41°N	139.31°E	Akita	39.44°N	140.08°E	454
5	Kurseong	26.55°N	88.19°E	Calcutta	22.27°N	88.18°E	497
6	Bracknell	52.03°N	1.13°W	Norddeich	53.34°N	7.07°E	585
7	Bracknell	52.03°N	1.13°W	Bockhacken	51.07°N	7.16°E	595
8	Allouis	47.00°N	2.00°E	Bockhacken	51.07°N	7.16°E	597
9	Oslo	59.26°N	10.36°E	Norddeich	53.34°N	7.07°E	686
10	Oslo	59.26°N	10.36°E	Luechow	52.59°N	11.13°E	718
11	Bracknell	52.03°N	1.13°W	Luechow	52.59°N	11.13°E	847
12	Xian	34.12°N	108.54°E	Beijing	39.57°N	116.27°E	925
13	Hyderabad	17.20°N	78.33°E	Trivandrum	8.29°N	76.59°E	999
14	Sanwa	36.11°N	139.51°E	Wakkanai	45.23°N	141.41°E	1,035
15	Beijing	39.57°N	116.27°E	Shanghai	31.15°N	121.29°E	1,069
16	Plymouth	50.25°N	4.49°W	Luechow	52.59°N	11.13°E	1,139
17	Xian	34.12°N	108.54°E	Shanghai	31.15°N	121.29°E	1,221
18	Derby	17.19°S	123.39°E	Alice spr.	23.32°S	133.41°E	1,253
19	Bombay	19.11°N	72.49°E	Trivandrum	8.29°N	76.59°E	1,272
20	Shannon	52.42°N	8.54°W	Luechow	52.59°N	11.13°E	1,347
21	Pori	61.28°N	21.52°E	Bockhacken	51.07°N	7.16°E	1,455
22	Ekala	7.06°N	79.54°E	Calcutta	22.27°N	88.18°E	1,930
23	Masirah	20.36°N	58.53°E	Delhi	28.43°N	77.12°E	2,056
24	Ankara	39.54°N	30.42°E	Chattonaye	46.46°N	6.57°E	2,057
25	Kavalla	40.58°N	24.21°E	Crowsley pk	51.31°N	0.57°W	2,255
26	Masirah	20.36°N	58.53°E	Trivandrum	8.29°N	76.59°E	2,365
27	Ankara	39.54°N	30.42°E	Jurbise	50.33°N	3.56°E	2,389
28	Ekala	7.06°N	79.54°E	Delhi	28.43°N	77.12°E	2,420
29	Poro	16.37°N	120.17°E	Beijing	39.57°N	116.27°E	2,621
30	Tinang	15.21°N	120.37°E	Beijing	39.57°N	116.27°E	2,765
31	Mahe	4.36°S	55.28°E	Trivandrum	8.29°N	76.59°E	2,795
32	Kranji	1.25°N	103.44°E	Calcutta	22.27°N	88.18°E	2,873
33	Kranji	1.25°N	103.44°E	Trivandrum	8.29°N	76.59°E	3,064
34	Jerusalem	32.04°N	34.47°E	Crowsley pk	51.31°N	0.57°W	3,611
35	Teheran	35.41°N	51.27°E	Luechow	52.59°N	11.13°E	3,665
36	Teheran	35.41°N	51.27°E	Bockhacken	51.07°N	7.16°E	3,886
37	Teheran	35.41°N	51.27°E	Norddeich	53.34°N	7.07°E	3,945
38	Kranji	1.25°N	103.44°E	Delhi	28.43°N	77.12°E	4,139
39	Kranji	1.25°N	103.44°E	Beijing	39.57°N	116.27°E	4,474
40	Sackville	45.53°N	64.19°W	Baldock	52.00°N	0.08°W	4,582
41	Kuwait	29.16°N	47.53°E	Crowsley pk	51.31°N	0.57°W	4,700
42	Ascension	7.54°S	14.23°W	Panorama	26.06°S	27.55°E	4,902
43	Sackville	45.53°N	64.19°W	Bockhacken	51.07°N	7.16°E	5,095

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Table 1
Continued

ID	TX	TX Geo. Lat. (°)	TX Geo. Lon. (°)	RX	RX Geo. Lat. (°)	RX Geo.Lon. (°)	Dist. (km)
44	Sackville	45.53°N	64.19°W	Chattonaye	46.46°N	6.57°E	5,277
45	New york	41.42°N	70.00°W	Norddeich	53.34°N	7.07°E	5,632
46	Sackville	45.53°N	64.19°W	Jokela	60.34°N	25.00°E	5,671
47	Kauai	22.00°N	159.48°W	Hiraiso	36.22°N	140.38°E	5,906
48	New York	41.42°N	70.00°W	Luechow	52.59°N	11.13°E	5,910
49	Darwin	12.25°S	130.37°E	Beijing	39.57°N	116.27°E	6,004
50	Greenville	35.35°N	77.22°W	Baldock	52.00°N	0.08°W	6,150
51	Greenville	35.35°N	77.22°W	Jurbise	50.33°N	3.56°E	6,468
52	Norfolk	36.48°N	76.30°W	Luechow	52.59°N	11.13°E	6,690
53	Daventry	52.15°N	1.08°W	Delhi	28.43°N	77.12°E	6,756
54	Greenville	35.35°N	77.22°W	Chattonaye	46.46°N	6.57°E	6,825
55	Carnarvon	24.54°S	113.43°E	Beijing	39.57°N	116.27°E	7,217
56	Greenville	35.35°N	77.22°W	Jokela	60.34°N	25.00°E	7,238
57	Wertachtal	48.05°N	10.41°E	Beijing	39.57°N	116.27°E	7,807
58	Ascension	7.54°S	14.23°W	Jokela	60.34°N	25.00°E	8,355
59	Shepparton	36.20°S	145.25°E	Trivandrum	8.29°N	76.59°E	8,689
60	Meyerton	26.35°S	28.08°E	Bockhacken	51.07°N	7.16°E	8,879
61	Tokyo	35.46°N	139.37°E	Norddeich	53.34°N	7.07°E	9,083
62	Fort collins	40.42°N	105.02°W	Hiraiso	36.22°N	140.38°E	9,144
63	Meyerton	26.35°S	28.08°E	Norddeich	53.34°N	7.07°E	9,145
64	Daventry	52.15°N	1.08°W	Panorama	26.06°S	27.55°E	9,159
65	Meyerton	26.35°S	28.08°E	Baldock	52.00°N	0.08°W	9,163
66	Mauritius	20.19°S	57.31°E	Luechow	52.59°N	11.13°E	9,287
67	Skelton	54.44°N	2.54°W	Panorama	26.06°S	27.55°E	9,459
68	Quito	0.14°S	78.20°W	Jurbise	50.33°N	3.56°E	9,483
69	Quito	0.14°S	78.20°W	Bockhacken	51.07°N	7.16°E	9,721
70	Delano	35.45°N	119.17°W	Beijing	39.57°N	116.27°E	9,849
71	Shepparton	36.20°S	145.25°E	Delhi	28.43°N	77.12°E	10,151
72	Quito	0.14°S	78.20°W	Jokela	60.34°N	25.00°E	10,754
73	Washington	39.00°N	76.48°W	Hiraiso	36.22°N	140.38°E	10,798
74	Darwin	12.25°S	130.37°E	Jokela	60.34°N	25.00°E	12,059
75	Carnarvon	24.54°S	113.43°E	Jokela	60.34°N	25.00°E	12,332
76	Koganei	35.41°N	139.31°E	Syowa	69.00°S	39.35°E	14,066
77	Sanwa	36.11°N	139.51°E	Syowa	69.00°S	39.35°E	14,129
78	Canberra	35.18°S	149.12°E	Luechow	52.59°N	11.13°E	16,206
79	Canberra	35.18°S	149.12°E	Norddeich	53.34°N	7.07°E	16,448

scenario and data requirements. The parameters are divided according to the following: local time h; frequency f; path distance D; mid-point geomagnetic latitude Φ; the twelve-month running mean value of the monthly sunspot number R_{12}; and season. The path midpoint used for geomagnetic-latitude classification was calculated as the midpoint along the short-path great-circle route between the transmitter and receiver. Specifically, the geographic coordinates of the transmitter and receiver were first converted into three-dimensional unit vectors on a spherical Earth. The two vectors were then averaged and normalized to obtain the geographic coordinates of the great-circle midpoint. Finally, the geographic midpoint was converted to geomagnetic coordinates, and its geomagnetic

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Table 2
Parameter Selection and Subset Classification Rules

Parameter	Subset classification
Path distance (D)	$0 < D < 1{,}000$ km; $1{,}000 \le D < 2{,}000$ km; $2{,}000 \le D < 5{,}000$ km; $5{,}000 \le D < 9{,}000$ km; $9{,}000 \le D < 12{,}000$ km; $12{,}000 \le D < 15{,}000$ km; $15{,}000 \le D < 18{,}000$ km
Local time at path midpoint (h)	h = 00:00, 01:00, …, 23:00
Season at path midpoint	Nov--Feb: NH winter/SH summer; Mar--Apr: NH spring/SH autumn; May--Aug: NH summer/SH winter; Sep--Oct: NH autumn/SH spring
Frequency (f)	$2 \le f \le 5$ MHz; $5 < f \le 10$ MHz; $10 < f \le 15$ MHz; $15 < f \le 30$ MHz
The twelve-month running mean value of the monthly sunspot number (R_{12})	$0 < R_{12} < 15$; $15 \le R_{12} < 45$; $45 \le R_{12} < 75$; $75 \le R_{12} < 105$; $105 \le R_{12} < 150$; $R_{12} \ge 150$
Geomagnetic latitude at path midpoint (Φ)	$0° \le \lvert\Phi\rvert \le 20°$; $20° < \lvert\Phi\rvert \le 40°$; $40° < \lvert\Phi\rvert \le 60°$; $\lvert\Phi\rvert > 60°$

Note. NH denotes the Northern Hemisphere, and SH denotes the Southern Hemisphere.

latitude was used as Φ for the latitude-based classification. The detailed rules for parameterization are shown in Table 2.

4. Analysis Results and Discussion

In this section, the field strength prediction performance of REC533, ICEPAC, and VOACAP is systematically evaluated using the internationally recognized CCIR data set. Following the standardized ITU assessment procedure, we examine the models' predictions across multiple dimensions. We further assess the models' performance from two complementary perspectives: (a) by comparing measured and predicted values along representative paths to evaluate their prediction ability, and (b) by analyzing field strength coverage maps to examine their spatial coverage characteristics.

4.1. Analysis Results

4.1.1. Overall Performance Analysis

In this section, all statistical comparisons are based on paired monthly median hourly field-strength values. Specifically, for each circuit, month, hour, frequency, and sunspot-number condition, the observed value is the monthly median of the hourly measured field strength, and the predicted value is the corresponding monthly median field strength calculated by REC533, ICEPAC, or VOACAP. The MD and SD values are then computed from the prediction-error samples $\varepsilon_i = p_i - m_i$ over all valid records or within each classified subset.

Figure 2 shows the frequency distributions of the prediction errors for REC533, ICEPAC, and VOACAP. The three panels correspond to REC533, ICEPAC, and VOACAP, respectively. The horizontal axis represents the field-strength prediction error, and the vertical axis represents the frequency of hourly prediction-error samples within each error interval. The MD and SD values are also shown in each panel to summarize the systematic bias

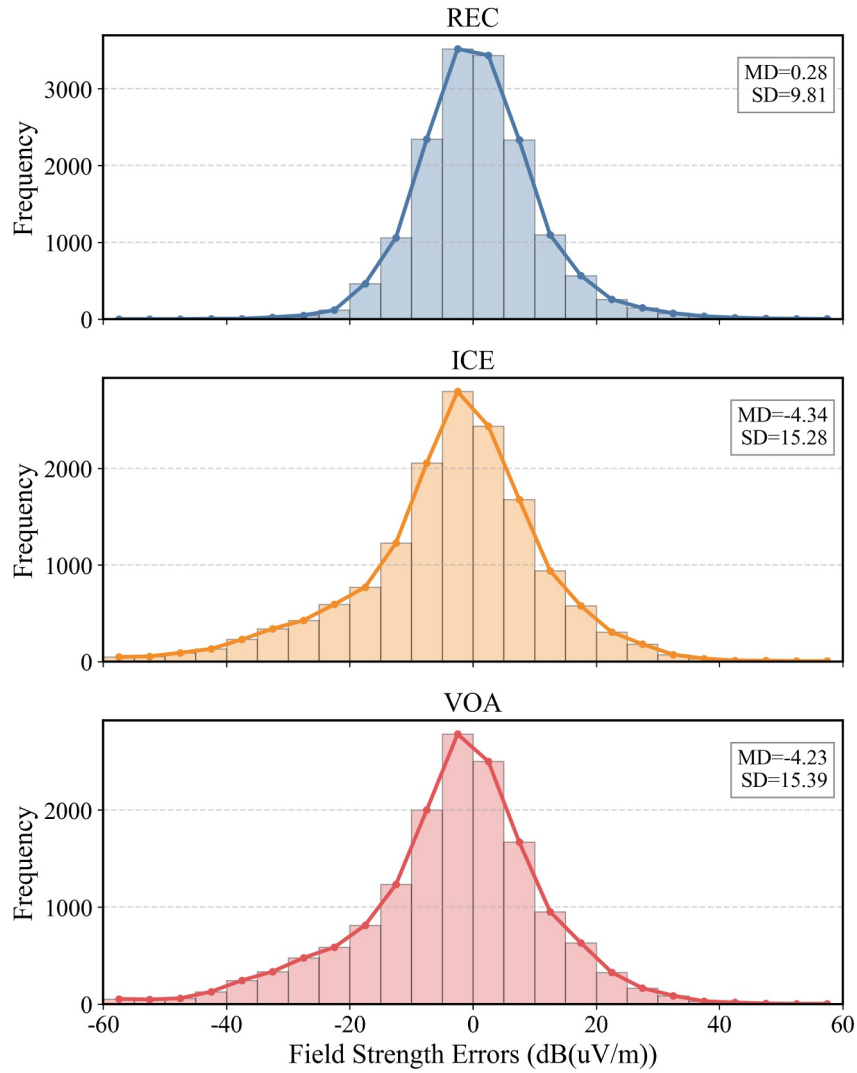


Figure 2. Frequency distributions of hourly prediction errors for REC533, ICEPAC, and VOACAP. The panels from top to bottom correspond to REC533, ICEPAC, and VOACAP, respectively. The *y*-axis represents the frequency of hourly prediction-error samples within each field-strength-error interval. MD and SD denote the mean difference and standard deviation of the prediction errors, respectively.

and dispersion of the prediction errors. REC533 exhibits the most concentrated error distribution, with an MD of 0.28 dB(μV/m) and an SD of 9.81 dB(μV/m), indicating an almost unbiased prediction and the lowest error dispersion among the three models. In contrast, ICEPAC and VOACAP show distributions shifted toward negative errors, with MD values of −4.34 dB(μV/m) and −4.23 dB(μV/m), respectively. Their SD values are also larger, reaching 15.28 dB(μV/m) and 15.39 dB(μV/m), respectively. These results indicate that ICEPAC and VOACAP systematically underestimate the received field strength and exhibit broader error distributions than REC533.

The prediction error of the median field strength at 24 local times of the receiving stations is given in Figure 3, where we can see that:

1. REC533 has the best overall prediction performance. It is smooth, and the difference from the measured value fluctuates around 0 dB(μV/m). The overall level of prediction is higher than that of the measured value between 09:00 and 18:00. ICEPAC and VOACAP fluctuate more, and the overall level of prediction is lower than that of the measured value except at 6–7 hr. At 11 o'clock, ICEPAC and VOACAP have the largest MDs relative to the observations, with values of −7 dB(μV/m) and −8.4 dB(μV/m), respectively.

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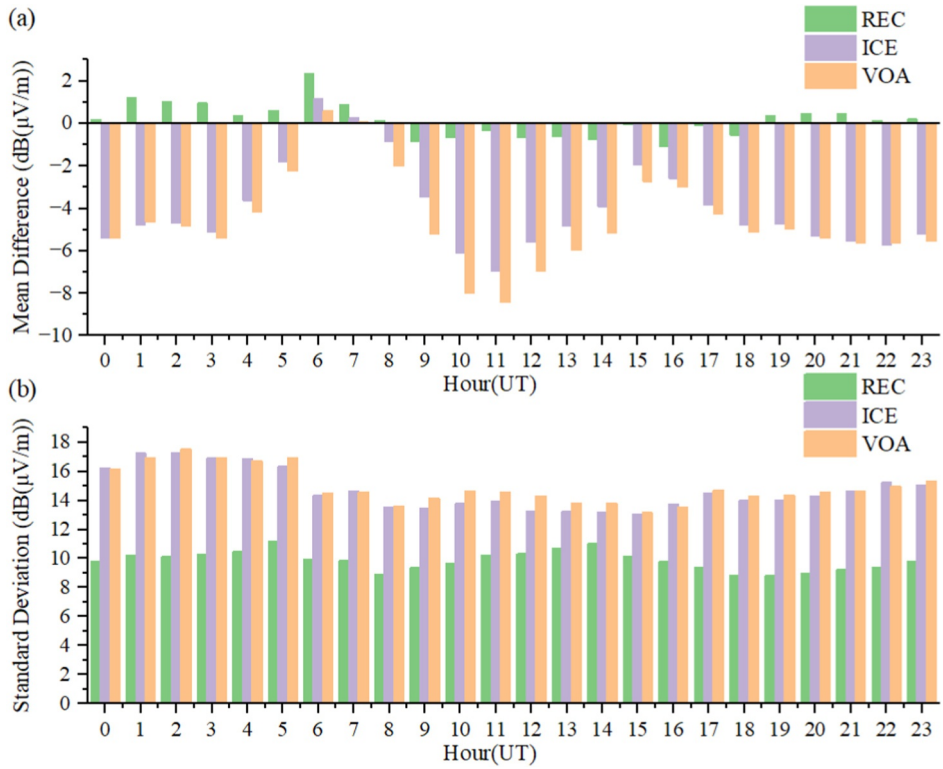


Figure 3. The predicted field strength errors of three models at 24 local times of the receiving stations.

2. REC533 exhibits consistently lower SD, indicating more stable predictions of the field strength across all hours. Peaks occur at 05:00 and 14:00, with values of 11.2 dB(μV/m) and 11 dB(μV/m), respectively. ICEPAC and VOACAP have higher SDs than REC533, with the most significant gaps at 01:00 (7.1 dB(μV/m) and 7.4 dB(μV/m)), indicating a broader prediction spread.

The prediction error of the median field strength within 7 distance ranges is given in Figure 4, where we can see that:

1. REC533's prediction is relatively smooth with distance. Excluding the 2,000–4,999 km and 9,000–11,999 km intervals, ICEPAC and VOACAP show an increasingly severe underestimation as the propagation distance increases.
2. REC533 shows the lowest prediction dispersion. ICEPAC and VOACAP have larger SD values than REC533, with the largest SDs appearing in the 15,000–17,999 km interval. ICEPAC and VOACAP exhibit similar overall prediction levels and dispersion characteristics.

The prediction error of the median field strength in 4 frequency bands is given in Figure 5, where we can see that:

1. The three models show similar prediction levels in the 5–10 MHz range. REC533 provides generally stable median field strength estimates with little variation across frequency. ICEPAC and VOACAP remain stable below 10 MHz, but their prediction spread increases markedly above 10 MHz. In the 10–15 MHz band, their MDs relative to the observations reach the largest values, at −10.6 dB (μV/m) and −11.1 dB (μV/m), respectively.
2. Among three models, the prediction of REC533 has a small degree of dispersion, while ICEPAC and VOACAP have a larger degree of overall dispersion in prediction. They show the largest differences from REC533 in the 10–15 MHz band, with SD gaps of 6.6 dB (μV/m) and 6.5 dB (μV/m), respectively.

The prediction error of the median field strength during 4 seasons is given in Figure 6, where we can see that:

1. The prediction of REC533 is generally stable, and it has the best prediction effect among the three models. ICEPAC and VOACAP have larger MDs for prediction, and their predictions are lower than the measured

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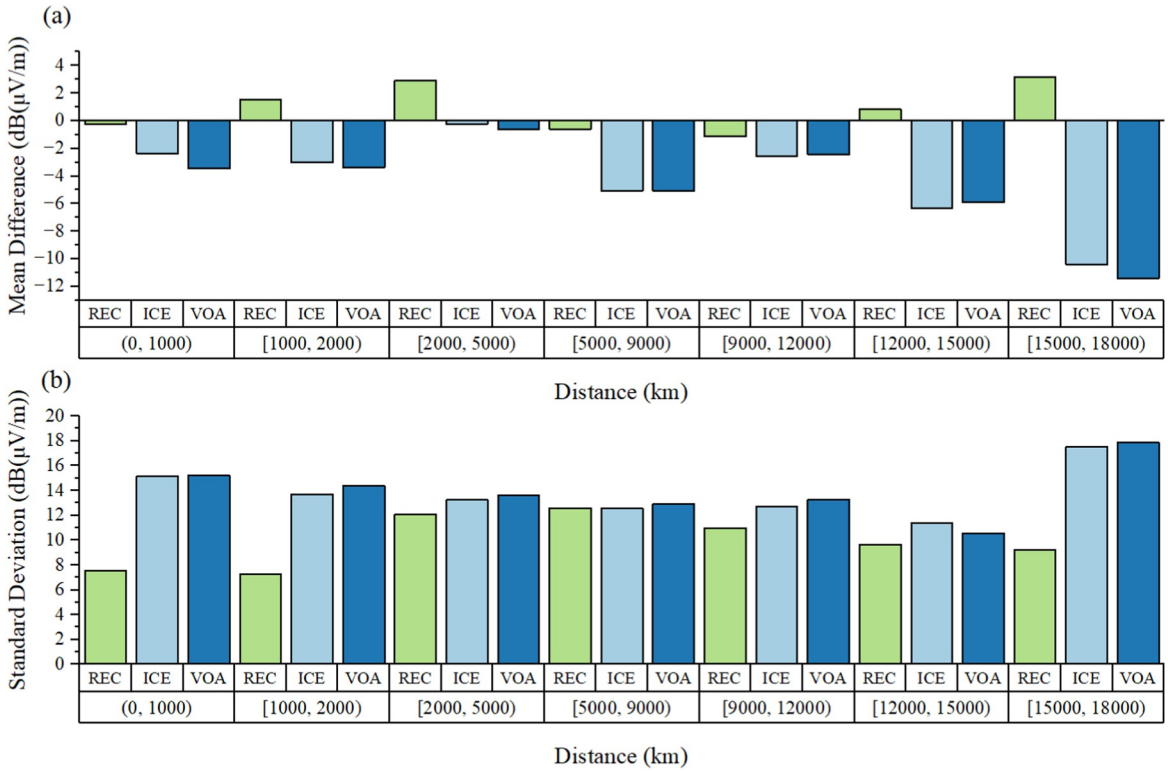


Figure 4. The predicted field strength errors of three models within 7 propagation distance ranges.

values. They have the largest MDs from the measured values in summer, with the values of −4.8 dB (μV/m) and −5.3 dB (μV/m), respectively.

2. REC533 exhibits the lowest dispersion, with its SD fluctuating around 10 dB (μV/m), indicating good stability in prediction. ICEPAC and VOACAP show noticeably higher dispersion, with the most significant differences from REC533 occurring in spring—5.3 dB (μV/m) and 5.7 dB (μV/m), respectively. The SDs of ICEPAC and VOACAP also differ the most in spring, with a maximum difference of 0.37 dB (μV/m).

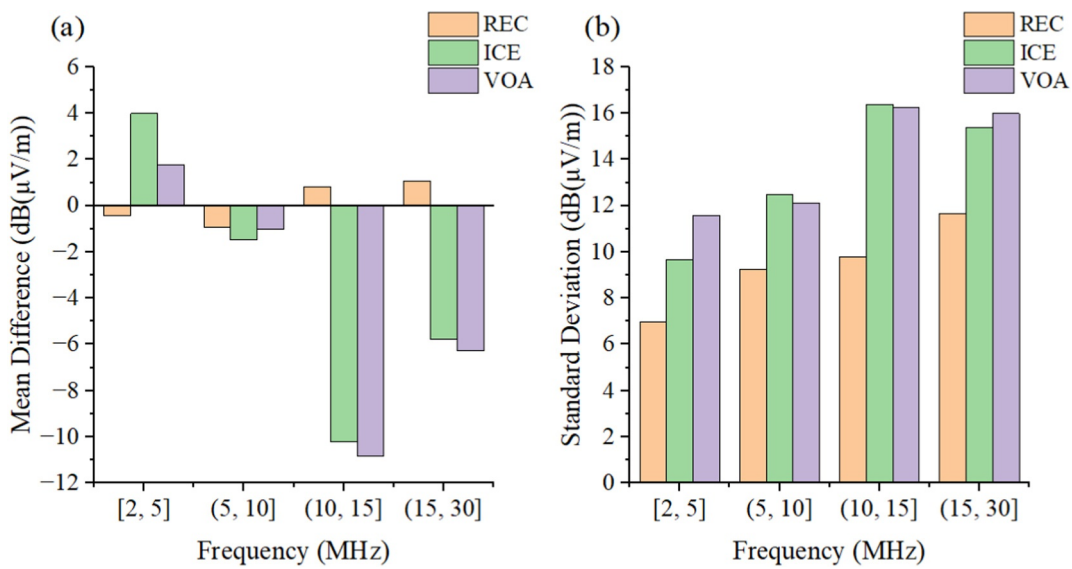


Figure 5. The predicted field strength errors in 4 frequency bands. (a) Mean Difference (MD); (b) Standard Deviation (SD).

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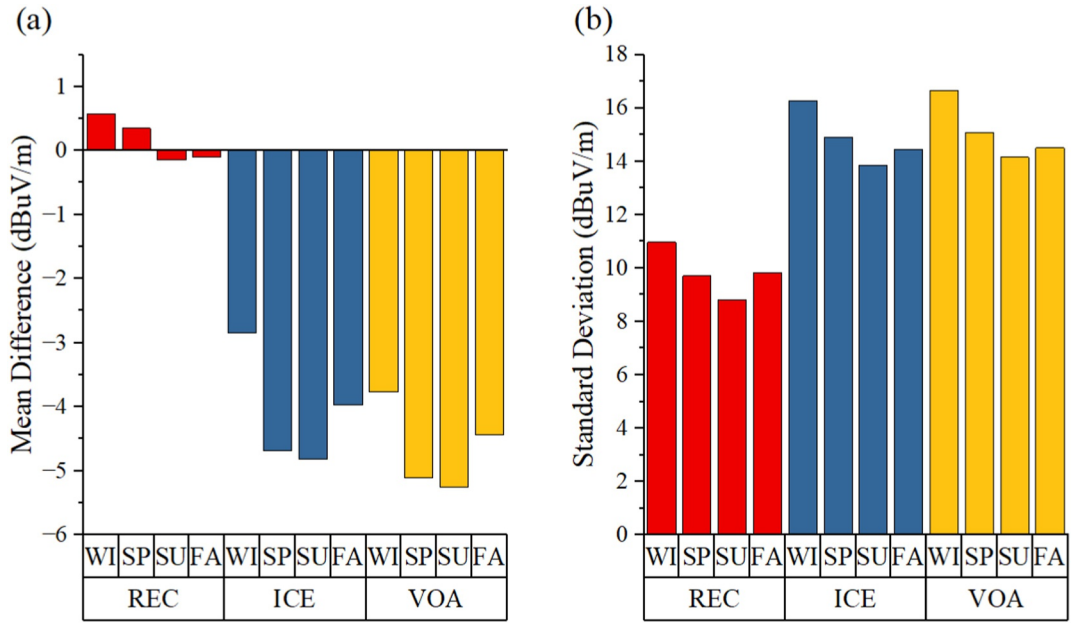


Figure 6. The prediction error of the median field strength during 4 seasons. WI represents winter, SP represents spring, SU represents summer, and FA represents fall.

The prediction error of the median field strength during 6 solar activity epochs is given in Figure 7, where we can see that:

1. REC533 shows relatively slight variations with sunspot number, indicating stable and accurate predictions of the field strength. ICEPAC and VOACAP consistently underestimate the field strength, with their predictions on average about 4 dB (μV/m) lower than those of REC533. The largest difference between ICEPAC and VOACAP occurs in the sunspot number range of 15–44, reaching 0.7 dB (μV/m).
2. The dispersion of REC533's predictions shows little dependence on sunspot number. ICEPAC and VOACAP exhibit much larger dispersion relative to REC533, with the maximum difference in SD reaching 7.5 dB (μV/m). And they exhibit more severe dispersion under low solar activity conditions.

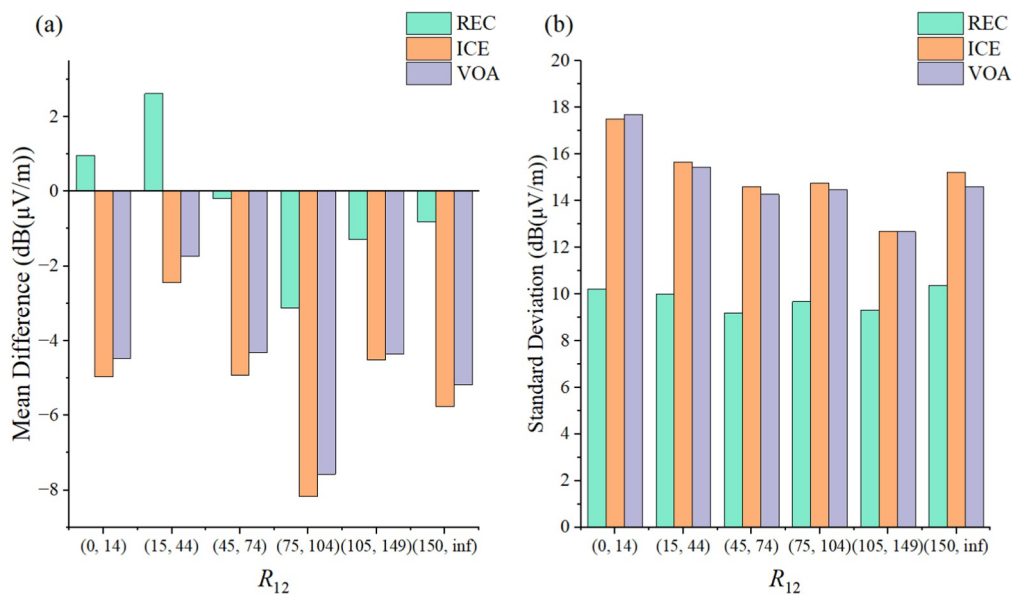


Figure 7. The predicted field strength errors of three models during 6 solar activity epochs.

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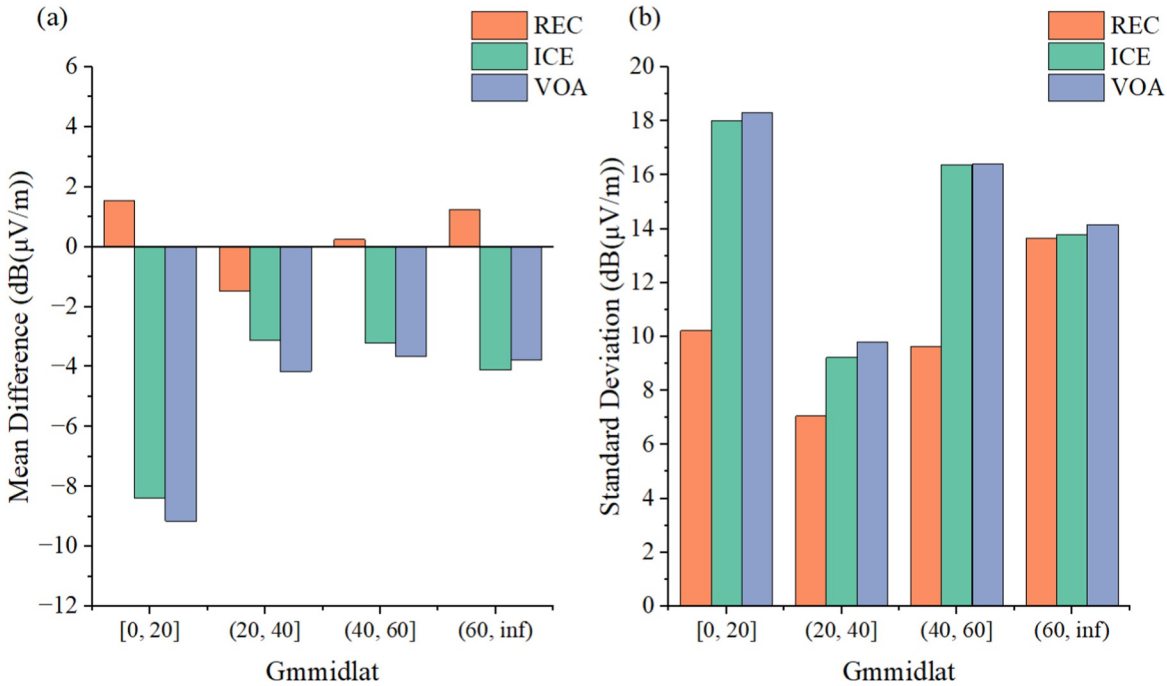


Figure 8. The predicted field strength errors of three models against mid-path geomagnetic latitude.

The prediction error of the median field strength as a function of mid-path geomagnetic latitude is given in Figure 8, where we can see that:

1. REC533 shows noticeable variations with geomagnetic latitude, predicting slightly higher than observations in the 20°–40° range, while slightly underestimating or closely matching the measurements at other latitudes. Overall, its deviations from the observations remain small. ICEPAC and VOACAP consistently underestimate the median field strength across all geomagnetic latitudes, with the largest differences from REC533 occurring in the 0°–20° range—9.9 dB(μV/m) and 10.7 dB(μV/m), respectively.
2. All three prediction models exhibit the smallest dispersion in the 20°–40° geomagnetic latitude range, and show comparable dispersion above 60°. At other latitudes, REC533 performs better, with significantly lower SD. The dispersion of ICEPAC and VOACAP is much larger than that of REC533 in the 0°–20° range, with differences in SD reaching 7.8 dB(μV/m) and 8.1 dB(μV/m), respectively. The SDs of ICEPAC and VOACAP differ the most within the 20°–40° range, with a maximum difference of 0.56 dB(μV/m).

4.1.2. Circuit-Specific Comparison Validation

To further illustrate the predictive performance of three models, this section compares the measured and predicted field strengths along representative propagation paths on an hourly basis. Line plots are generated to examine the models' fitting accuracy and their differences during diurnal variations.

We selected twelve circuits for comparison. In this study, a circuit refers to a specific transmitter–receiver path under a given observation condition, including month, season, and solar activity level. Their selection follows the principle that the path-midpoint geomagnetic latitude λ of the circuits should cover three typical regions: low latitude ($|\lambda| \leq 20°$), mid-latitude ($20° < |\lambda| < 60°$), and high-latitude ($|\lambda| \geq 60°$). Here, λ refers to the geomagnetic latitude of the midpoint of the transmitter–receiver path, rather than the geographic latitude of either station. This selection ensures that the analysis captures the differences in propagation characteristics under different ionospheric structures (International Telecommunication Union, 1998). The geomagnetic latitudes of the circuits are distributed as evenly as possible across the latitude bands to ensure a reasonable spatial distribution of the samples. To investigate the effect of solar activity intensity on model prediction performance, our study follows the research method of Huth et al. (Huth & Stryhal, 2024). We count the sunspot numbers in December from 1964 to 1985. Months in the lower 25% of the sunspot-number distribution are classified as low solar activity, and those

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Table 3
Information on the Twelve Selected Typical Circuits

ID	Test year	Month	Solar activity level	Transmitter	Receiver	Path midpoint latitude (°)	Midpoint ionosphere region	Season
143	1975	4	Low	Mauritius	Luechow	16.20°N	Low latitude	Autumn
143	1975	7	Low	Mauritius	Luechow	16.20°N	Low latitude	Winter
163	1980	1	High	Canberra	Luechow	8.71°N	Low latitude	Summer
164	1979	10	High	Canberra	Luechow	8.71°N	Low latitude	Spring
3	1984	10	Low	Sanwa	Akita	37.77°N	Middle latitude	Autumn
4	1980	4	High	Sanwa	Akita	37.77°N	Middle latitude	Spring
4	1980	7	High	Sanwa	Akita	37.77°N	Middle latitude	Summer
5	1975	1	Low	Koganei	Akita	37.42°N	Middle latitude	Winter
96	1985	10	Low	New York	Norddeich	47.38°N	High latitude	Autumn
104	1980	1	High	New York	Luechow	47.01°N	High latitude	Winter
105	1980	4	High	New York	Luechow	47.01°N	High latitude	Spring
114	1977	7	Low	Norfolk	Luechow	44.53°N	High latitude	Summer

Note. The ID corresponds to the path ID listed in Table 1. The path-midpoint latitude refers to the geomagnetic latitude of the midpoint along the short-path great-circle route.

in the upper 25% as high solar activity. To analyze seasonal differences, four circuits are selected for each latitude region, corresponding to spring, summer, autumn, and winter. This allows multi-dimensional comparative validation across different latitudes, seasons, and solar-activity conditions. The definitions of circuit geomagnetic latitude, time, and season are the same as Section 3.2. The season of each selected circuit was determined according to the observation month and the hemisphere of the path midpoint. For the Northern Hemisphere, November–February, March–April, May–August, and September–October were classified as winter, spring, summer, and autumn, respectively, while the corresponding opposite seasons were used for the Southern Hemisphere.

The result of the circuits selected by the above rule is shown in Table 3. The twelve circuits used in this study are distributed across seven paths: Sanwa–Akita, Koganei–Akita, New York–Norddeich, New York–Luechow, Norfolk–Luechow, Mauritius–Luechow, and Canberra–Luechow. The twelve selected typical circuits represent transmitter–receiver paths whose path-midpoint geomagnetic latitudes fall into low-, mid-, and high-latitude categories. Each regional circuit corresponds to the four seasons of spring, summer, fall, and winter, as well as two periods of high and low solar activity. Figure 9 shows the circuit-specific validation results for the twelve selected typical circuits listed in Table 3. The figure is arranged according to latitude region and season. The three rows correspond to low-, middle-, and high-latitude regions, and the four columns correspond to winter, spring, summer, and autumn. In each panel, OBS represents the observed monthly median hourly field strength from the CCIR data set, while REC, ICE, and VOA represent the corresponding predictions from REC533, ICEPAC, and VOACAP. The horizontal axis denotes local time, and the vertical axis denotes the received field strength in dB (μV/m).

For the low-latitude circuits, the observed field strength exhibits clear diurnal variation, and model performance varies across seasons. In the winter case, the three models generally reproduce the variation trend after the early-morning period, but ICEPAC and VOACAP show larger deviations at several hours. In the spring and summer cases, the available observations are mainly concentrated in nighttime or daytime intervals, and REC533 generally follows the observed values more closely. ICEPAC and VOACAP show more pronounced deviations in some daytime periods. In the autumn case, all three models capture the decrease in field strength around local daytime and the recovery during evening hours, while REC533 remains closer to the observations for most hours.

For the middle-latitude circuits, the diurnal variations are smoother and more consistent among the models. In all four seasonal cases, the observed field strength decreases during daytime and recovers during evening and nighttime. REC533 generally agrees well with the observations and reproduces both the daytime minimum and the nighttime recovery. ICEPAC and VOACAP also capture the overall diurnal pattern, but they tend to produce

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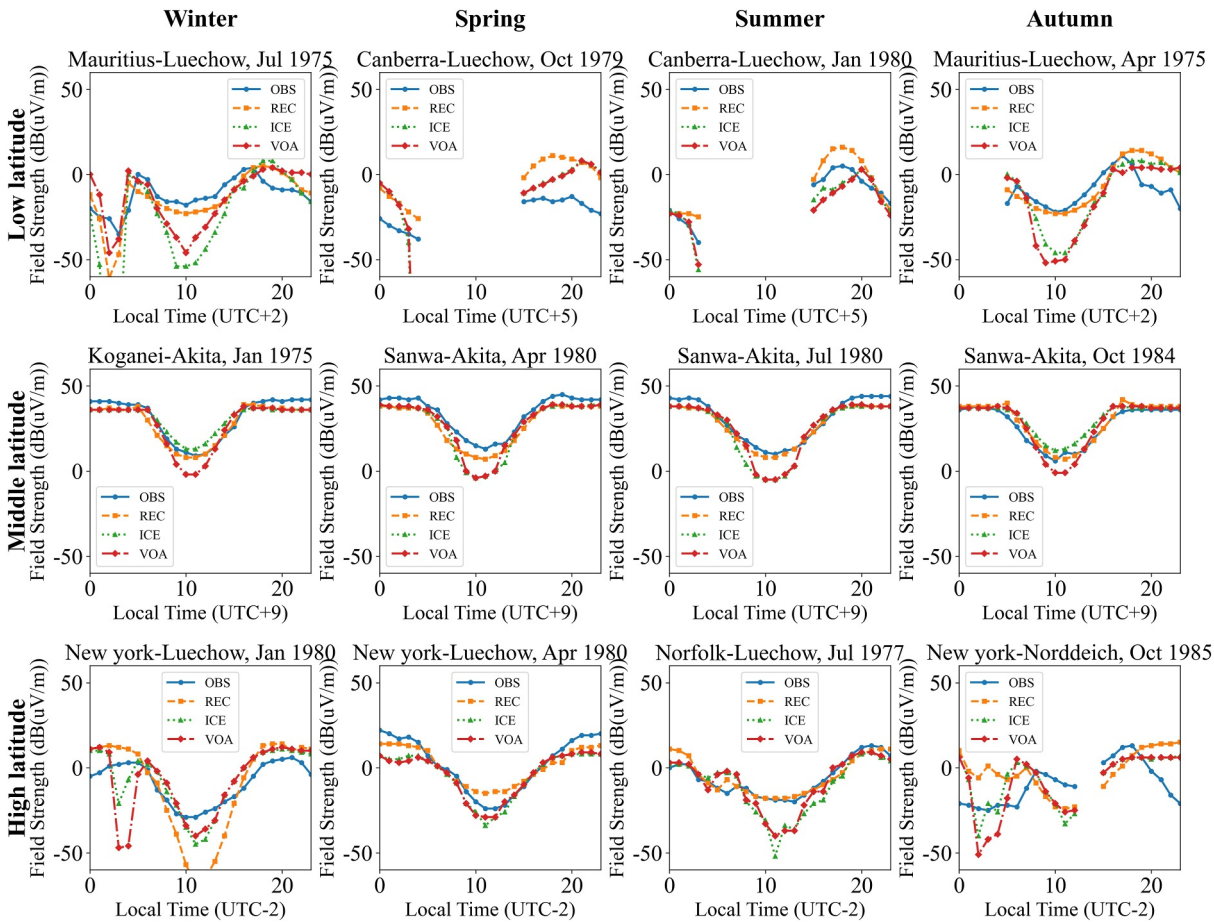


Figure 9. Diurnal variations of the observed and predicted median field strength for twelve representative circuits. Rows correspond to low-, middle-, and high-latitude regions, and columns correspond to winter, spring, summer, and autumn.

lower daytime field-strength values in several panels. Among the twelve circuits, the middle-latitude cases show the most consistent agreement between the observations and model predictions.

For the high-latitude circuits, the temporal variations are more complex than those in the middle-latitude region. In the winter case, the model curves differ substantially from the observations during several hours, and ICEPAC and VOACAP show strong short-term fluctuations. In the spring and summer cases, the three models reproduce the general decrease in field strength during local daytime and the recovery during nighttime, but ICEPAC and VOACAP show larger daytime underestimation in some periods. In the autumn case, observations are available only during part of the day, and the predicted curves show larger deviations from the measured values. REC533 generally follows the observed trend more closely, whereas ICEPAC and VOACAP exhibit larger temporal deviations.

In summary, the path-specific results align with the overall performance analysis, confirming the reliability of the earlier conclusions. REC533 provides the most accurate and stable predictions across all latitude regions, with diurnal variations closely matching the observations. ICEPAC and VOACAP show daytime underestimation or fluctuations at low and high latitudes, with ICEPAC performing better than VOACAP at high latitudes during nighttime.

4.1.3. Field Strength Coverage Analysis

To illustrate the field strength prediction characteristics of three models across different regions intuitively, six transmitting stations from six continents were selected: Canberra (35.3°S, 149.1°E) in Oceania, Fort Collins (40.6°N, 105.1°W) in North America, Meyerton (26.6°S, 28.0°E) in Africa, Oslo (59.9°N, 10.8°E) in Europe,

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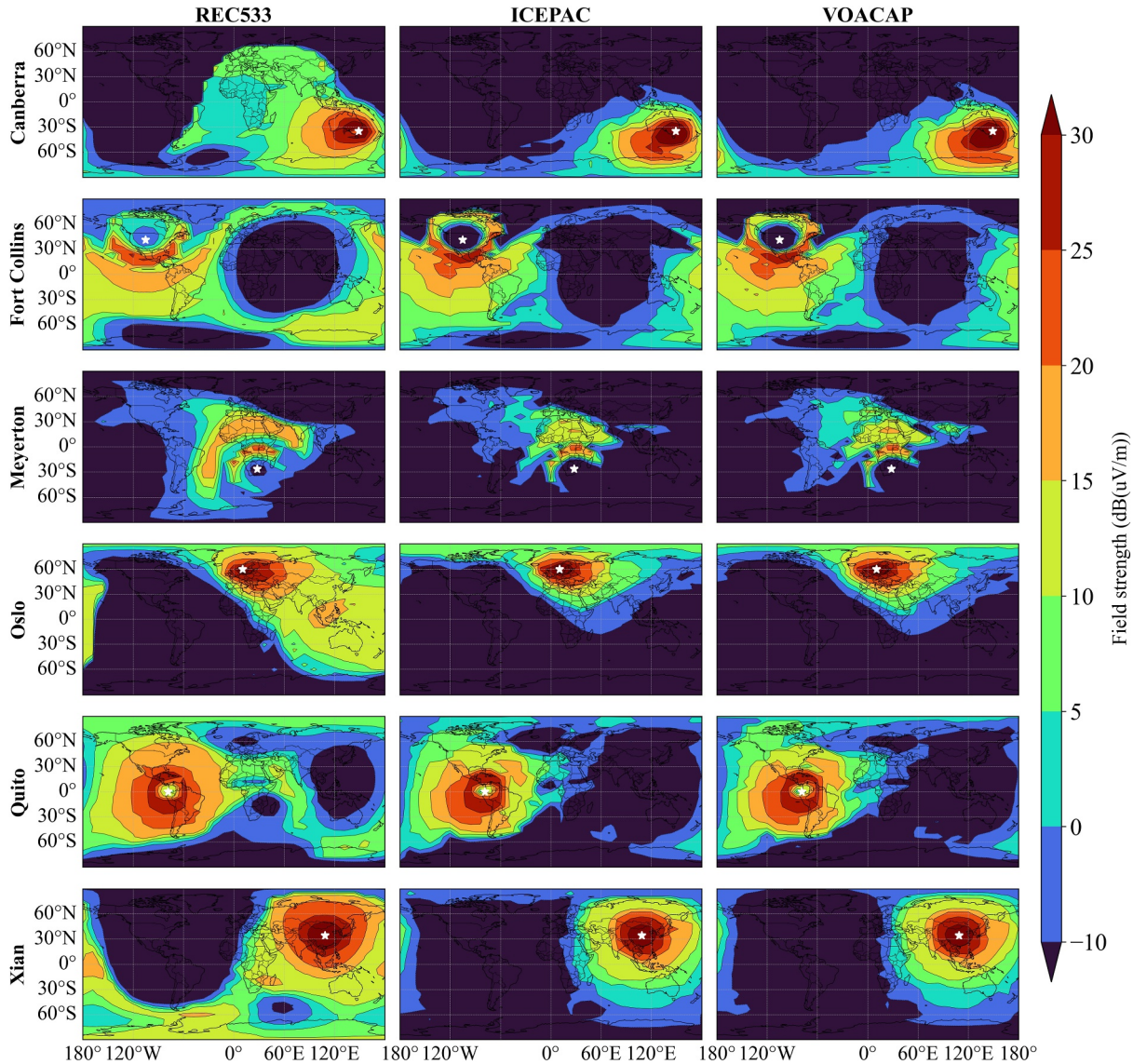


Figure 10. Field-strength coverage maps predicted by REC533, ICEPAC, and VOACAP for six transmitting stations. Rows correspond to Canberra, Fort Collins, Meyerton, Oslo, Quito, and Xian, respectively, and columns correspond to REC533, ICEPAC, and VOACAP, respectively. The white star denotes the transmitting station, and the color scale represents the predicted field strength in dB(μV/m).

Quito (0.2°S, 78.5°W) in South America, and Xian (34.3°N, 108.9°E) in Asia. These stations span both hemispheres and cover low-, mid-, and high-latitude regions, enabling a broad comparison of coverage behavior. For each station, the observation record corresponding to the maximum measured field strength in the CCIR data set was used to determine the input parameters, including month, operating frequency, sunspot number, and hour. The selected input conditions were Canberra (Month 10, Frequency 6.4 MHz, SSN 96, Hour 17 UT), Fort Collins (Month 7, Frequency 7.3 MHz, SSN 140, Hour 15 UT), Meyerton (Month 10, Frequency 21.5 MHz, SSN 29, Hour 18 UT), Oslo (Month 10, Frequency 6.4 MHz, SSN 96, Hour 17 UT), Quito (Month 8, Frequency 11.8 MHz, SSN 40, Hour 3 UT), and Xian (Month 7, Frequency 7.3 MHz, SSN 140, Hour 15 UT). These input conditions were kept the same for REC533, ICEPAC, and VOACAP in each station case. Figure 10 shows the field-strength coverage maps predicted by the three models. Each row corresponds to one transmitting station, and each column corresponds to one prediction model. The white star denotes the location of the transmitting station, and the color scale represents the predicted field strength in dB(μV/m). The coverage maps are model-predicted field-strength distributions. These coverage maps, generated under these representative conditions, reveal the spatial

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distribution patterns of three prediction models and provide a direct basis for comparing their coverage and attenuation characteristics. The coverage characteristics of the six transmitting stations are analyzed individually below.

For the Canberra case, REC533 shows the widest spatial coverage among the three models. The 10–20 dB(μV/m) field-strength region extends over a broad area from Australia toward the Indian Ocean and parts of Asia. ICEPAC and VOACAP show more limited coverage, with high field-strength values mainly concentrated around Australia and nearby oceanic regions. Compared with REC533, their low-to-moderate field-strength regions shrink more rapidly with distance.

For the Fort Collins case, all three models predict strong field-strength values around North America. REC533 shows broader 10–20 dB(μV/m) coverage extending toward the North Atlantic, South America, Europe, and North Africa. ICEPAC and VOACAP predict stronger field strength mainly over North America and adjacent regions, while the distant coverage area is smaller than that of REC533.

For the Meyerton case, REC533 produces a relatively broad coverage pattern extending from southern Africa toward central Africa and the Indian Ocean. ICEPAC and VOACAP predict smaller coverage areas, with the stronger field-strength regions mainly concentrated near southern Africa and adjacent coastal regions. The field strength decreases rapidly over distant oceanic and continental areas in the ICEPAC and VOACAP results.

For the Oslo case, REC533 predicts an extensive coverage region over Europe, western Russia, and surrounding areas. The 10–20 dB(μV/m) region extends farther southeastward than in the other two models. ICEPAC and VOACAP show more concentrated coverage around northern and central Europe, with weaker field strength over distant regions.

For the Quito case, REC533 shows broad coverage over South America and the eastern Atlantic region. ICEPAC and VOACAP also predict strong field strength around the transmitter, but their coverage is more limited eastward. The field-strength values over the Atlantic and African regions are generally lower than those predicted by REC533.

For the Xian case, REC533 predicts wide coverage over East Asia, Central Asia, Southeast Asia, and parts of the western Pacific. ICEPAC and VOACAP show smaller coverage areas, with stronger field-strength values mainly concentrated around China and nearby regions. Compared with REC533, their 10–20 dB(μV/m) coverage regions are more limited.

Overall, the coverage results from the six transmitting stations reveal a consistent pattern among the three models. REC533 maintains higher field strengths over the most significant areas, with the 20–30 dB(μV/m) and 10–20 dB (μV/m) intervals extending the farthest and showing the slowest attenuation. In contrast, ICEPAC and VOACAP provide more limited coverage, with their stronger signals confined to smaller regions and the 10–20 dB(μV/m) levels diminishing rapidly with distance. Comparisons indicate that REC533 provides the most accurate predictions. Therefore, this model is recommended for engineering applications.

4.2. Discussion

1. REC533 achieves the best overall performance, exhibiting a near-zero bias of 0.28 dB(μV/m) and the lowest dispersion of 9.81 dB(μV/m), together with consistently high stability across all propagation scenarios. In contrast, ICEPAC and VOACAP exhibit systematic underestimation of approximately 4 dB(μV/m), with substantially larger dispersion exceeding 15 dB(μV/m), particularly for long-distance circuits, at high frequencies, and in low-latitude regions. This difference may be related to the different model structures and their treatments of both short- and long-distance propagation. REC533 uses a standardized ITU-R control-point procedure with explicitly separated loss terms and a composite-mode long-distance formulation (International Telecommunication Union, 2019), whereas the IONCAP-derived models rely more strongly on electron-density-profile-based ray-hop calculations, mode selection, and different long-distance treatments (Stewart, 2008).
2. Hourly comparisons on twelve representative circuits show that the three prediction models reproduce the general diurnal variation. REC533 agrees most closely with observations across all latitudes. ICEPAC and VOACAP exhibit notable daytime underestimation at low and high latitudes. ICEPAC performs slightly better than VOACAP at high latitudes during nighttime. These results confirm the robustness of REC533 and reveal

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scenario-dependent limitations in the other two models. These differences are consistent with their model-specific ionospheric treatments. ICEPAC introduces ICED-based electron-density and conductivity profiles, Chapman-layer representation, and high-latitude regional corrections, whereas VOACAP largely retains the empirical IONCAP signal-power framework (Lane, 2005).

3. Spatial coverage analysis across six transmitting stations shows that REC533 attenuates most slowly and maintains the broadest 20–30 dB(μV/m) and 10–20 dB(μV/m) coverage. ICEPAC and VOACAP exhibit much smaller coverage and faster decay, especially over oceanic and high-latitude regions. These results agree with the statistical and circuit-level analyses and further confirm the superior accuracy and spatial consistency of REC533. The broader and smoother REC533 coverage may be partly associated with its standardized loss terms and long-distance composite-mode treatment (International Telecommunication Union, 2019), while the faster attenuation and more limited coverage of ICEPAC and VOACAP may reflect the sensitivity of the IONCAP-derived framework to electron-density-profile assumptions, mode selection, and long-distance propagation treatment (Lane, 2001; Stewart, 2008).

5. Conclusions

This study systematically compares the performance of the three field-strength prediction models, REC533, ICEPAC, and VOACAP, using the CCIR measurement data set and the ITU standardized evaluation method. Through multi-dimensional analysis and three complementary validation experiments, the study evaluates their accuracy, applicability, and limitations.

Based on the statistical, circuit-level, and spatial-coverage results, REC533 is recommended as the primary model for HF circuit planning, frequency allocation, and reliability analysis. Its stable performance under diverse ionospheric and propagation conditions supports global-scale engineering use. ICEPAC and VOACAP remain useful for short- and medium-range paths and mid-frequency bands, but should be applied with caution under long-distance, low-latitude daytime, or high-solar-activity conditions. The results provide quantitative guidance for HF frequency planning, circuit design, and emergency communication support. Future work may incorporate machine learning techniques, such as data-driven calibration of ionospheric models, to further improve prediction accuracy under high-latitude, long-distance, and extreme solar activity conditions.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The data used in this work and the produced data files of the proposed model are available (Zhu et al., 2025). The implementations of REC533, ICEPAC, and VOACAP used in this study were obtained from the official website of the Institute for Telecommunication Sciences, National Telecommunications and Information Administration (NTIA/ITS): https://its.ntia.gov/software/high-frequency/windows-32-bit-hf-propagation-models. The NTIA/ITS HF propagation software package includes the Windows implementations of REC533, ICEPAC, and VOACAP. The area-coverage field-strength data used for Figure 10 were generated using the AREAWIN area-coverage program included in this software package. For each case, the transmitter coordinates, month, operating frequency, sunspot number, and hour were specified as input conditions, as listed in the corresponding figure captions. The generated model-output data used to produce Figure 10 are also included (Zhu et al., 2025).

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